NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND

MIL-PRF-8805/93C
29 March 1975
SUPERSEDING
MIL-S-8805/93B
18 May 1984

PERFORMANCE SPECIFICATION SHEET

SWITCHES, TOGGLE, UNSEALED, 7.0 AMPERES

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-8805.

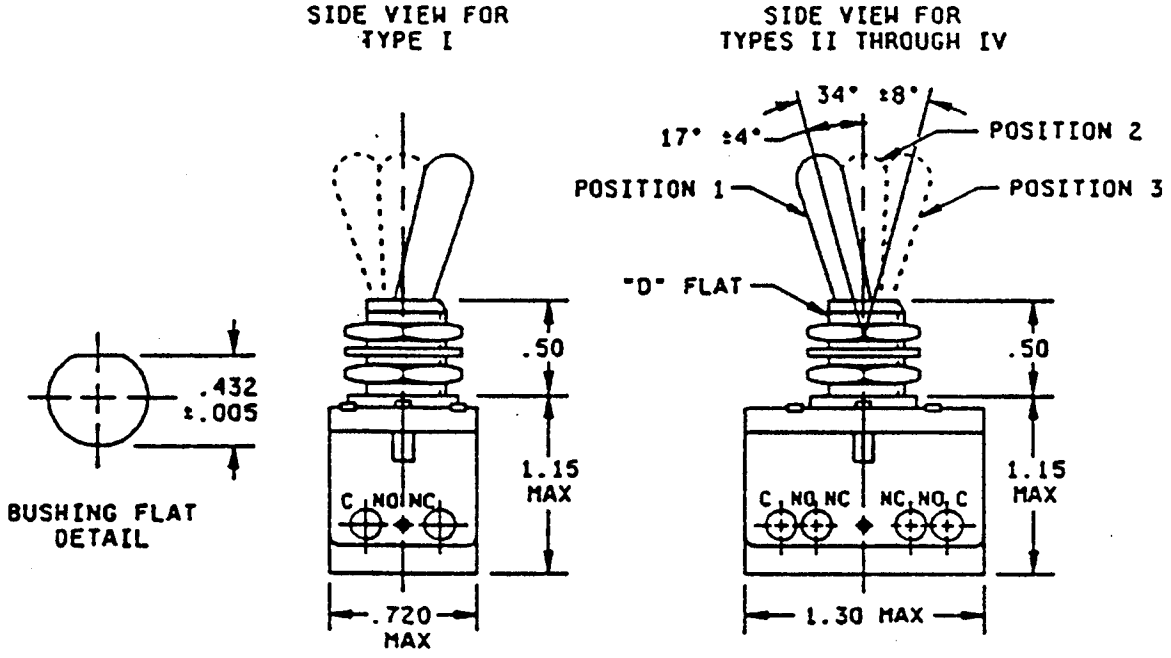
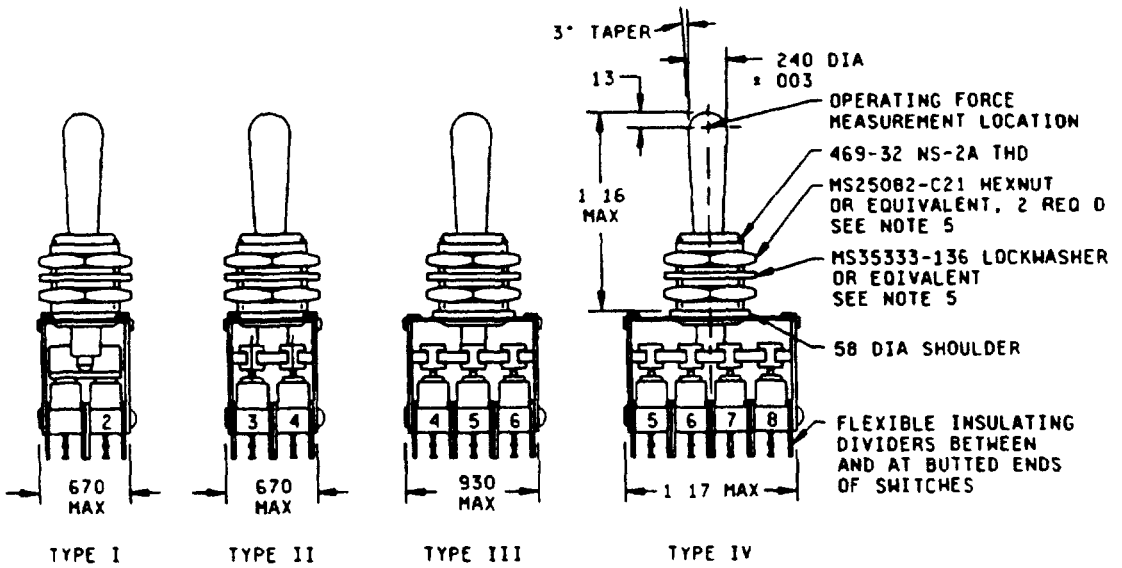


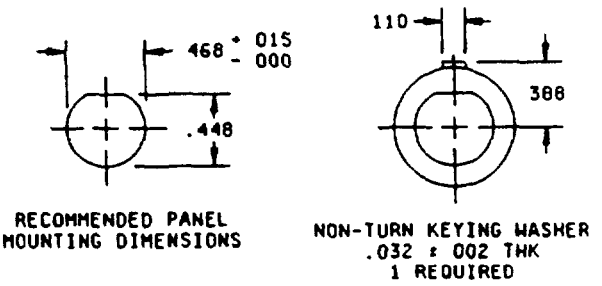
FIGURE 1. Configurations and dimensions.

© denotes changes

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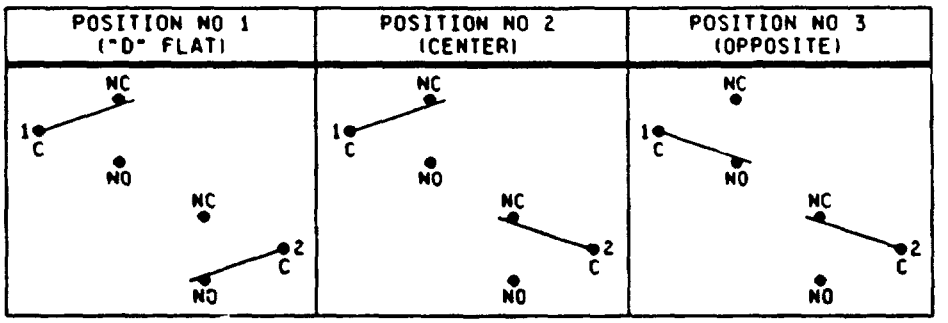
Inches	mm	Inches	mm	Inches	mm	Inches	mm
.002	0 05	.110	2.79	.468	11.89	.930	23.62
.003	0 08	.13	3.30	.469	11.91	1.15	29.21
.005	0.13	.240	6.10	.50	12.70	1.16	29.46
.015	0.38	.388	9.86	.56	14.22	1.17	29.72
.03	0.76	.432	10.97	.670	17.02	1.30	33.02
.032	0.81	.448	11.38	.720	18.29		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.03 inch (0.76 mm) for two place decimals and ±.005 inch (0.13 mm) for three place decimals.
4 Configuration optional provided that specified envelope dimensions are not exceeded.
5. Alternative base metals and protective finishes, as approved by the qualifying activity, may be utilized for hexagon nut and lock washer material. Dimensions shall be in accordance with the referenced hardware specifications.

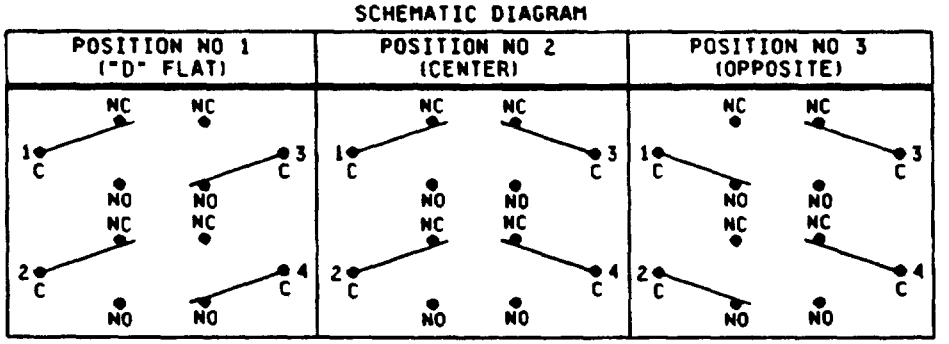
FIGURE 1 Configurations and dimensions Continued

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TYPE I
SCHEMATIC DIAGRAM



TYPE II
SCHEMATIC DIAGRAM



TYPE III
SCHEMATIC DIAGRAM

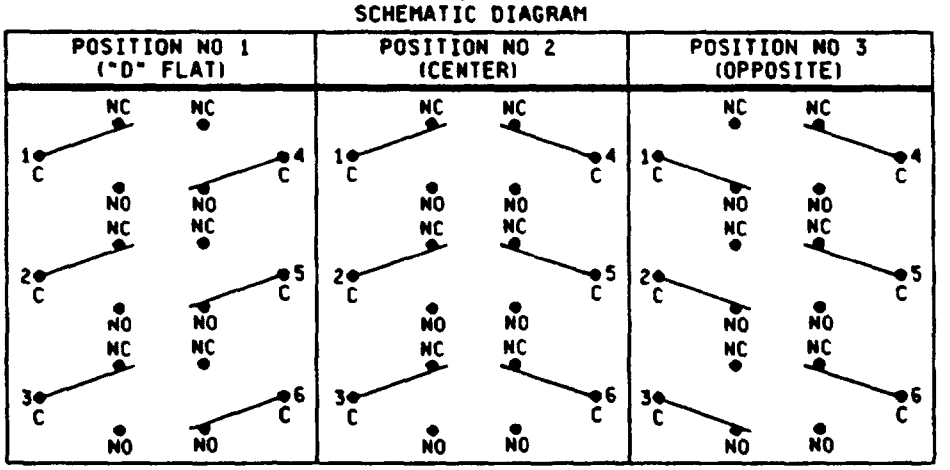


FIGURE 2 Schematic diagrams

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TYPE IV
SCHEMATIC DIAGRAM

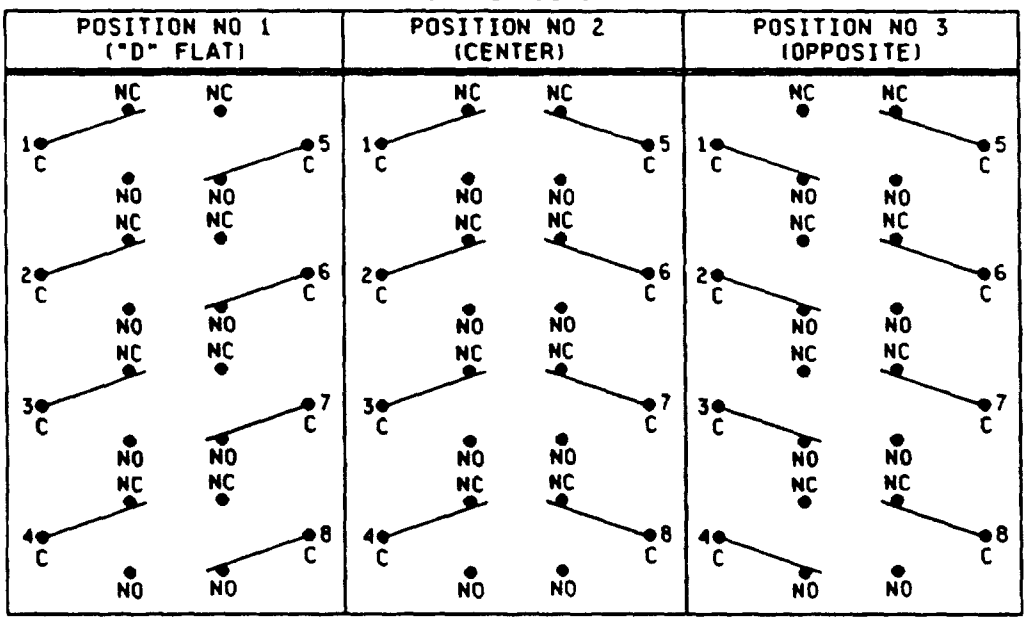


FIGURE 2 Schematic diagrams - Continued

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REQUIREMENTS

Dimensions and configuration See figure 1 Basic switch shall be MS24547-4 (MIL-S-8805/4)

Enclosure design: 1 (unsealed). Use boot in accordance with MIL-B-5423/2 (PIN M5423/02-01) for panel seal.

Temperature characteristic: 1 (-55°C to +85°C).

Shock (specified pulse): MIL-STD-202, method 213, test condition C, (100 g, half sine).

Sinusoidal vibration grade: 1 (10 to 500 Hz, 10 g peak).

Weight:

Type I: 0.078 pound maximum.

Types II and III: 0.093 pound maximum.

Type IV: 0.125 pound maximum.

Terminal strength: 5 pounds.

Contact resistance: Initial 50 milliohms, 100 milliohms after mechanical endurance.

Operating characteristics:

Coincidence of operating and release points: All poles shall transfer within 5° of lever travel.

Operating force: 6 pounds maximum.

Material:

Lever and bracket: Corrosion-resistant steel

(C) Dielectric withstanding voltage at reduced barometric pressure:

Altitude: 50,000 feet (MIL-STD-202, method 105, test condition B).

Mechanical endurance: 50,000 cycles at a cycling rate of 15 to 65 cycles per minute

Electrical endurance: 25,000 cycles

Electrical ratings: See table I

Part or Identifying Number (PIN): See table II.

Qualification:

Group submission: See table III.

(C) Group B inspection: Subject one M8805/93-002 or -003, and one -027 or -028 to each of the test groups specified in table IV.

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TABLE I. Electrical ratings.

Loads	Sea level		50,000 feet
	28 V dc	115 V ac, 60 Hz	28 V dc
	(amperes)	(amperes)	(amperes)
Resistive	7.0	7.0	4.0
Inductive	4.0	7.0	2.5
Lamp	2 5	2 0	2.5

TABLE II. PIN's and characteristics.

PIN M8805/93	Type (see figure 2)	Circuit	Toggle lever operation in "D" flat side - center - side opposite "D" flat
-001	I	2PDT	Maintained - maintained - maintained
-002	I	2PDT	Momentary - maintained - maintained
-003	I	2PDT	Maintained - maintained - momentary
-004	I	2PDT	Momentary - maintained - momentary
-005	I	2PDT	Maintained - none - maintained
-006	I	2PDT	Momentary - none - maintained
-007	I	2PDT	Maintained - none - momentary
-008	II	4PDT	Maintained - maintained - maintained
-009	II	4PDT	Momentary - maintained - maintained
-010	II	4PDT	Maintained - maintained - momentary
-011	II	4PDT	Momentary - maintained - momentary
-012	II	4PDT	Maintained - none - maintained
-013	II	4PDT	Momentary - none - maintained
-014	II	4PDT	Maintained - none - momentary
-015	III	6PDT	Maintained - maintained - maintained
-016	III	6PDT	Momentary - maintained - maintained
-017	III	6PDT	Maintained - maintained - momentary
-018	III	6PDT	Momentary - maintained - momentary
-019	III	6PDT	Maintained - none - maintained
-020	III	6PDT	Momentary - none - maintained
-021	III	6PDT	Maintained - none - momentary
-022	IV	8PDT	Maintained - maintained - maintained
-023	IV	8PDT	Momentary - maintained - maintained
-024	IV	8PDT	Maintained - maintained - momentary
-025	IV	8PDT	Momentary - maintained - momentary
-026	IV	8PDT	Maintained - none - maintained
-027	IV	8PDT	Momentary - none - maintained
-028	IV	8PDT	Maintained - none - momentary

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TABLE III Qualification inspection.

Examination or test	Test sample	Extent of approval
Group I (14 sample units) Visual and mechanical examination Operating characteristics	All sample units for groups II through V, plus 1 each of any type II unit and any type III unit	
Group II (4 sample units) Strength of actuating means Strength of mounting bushing Thermal shock Vibration Shock (specified pulse) Operating characteristics Dielectric withstanding voltage Visual and mechanical examination Marking visibility	2 sample units of each of the following: M8805/93-002 or -003 and M8805/93-027 or -028	All PIN's
Group III (2 sample units) Salt spray (corrosion) Visual and mechanical examination Marking visibility	1 sample unit of each of the following: M8805/93-002 or -003 and M8805/93-027 or -028	
Group IV (4 sample units) Contact resistance Low temperature operation 1/ Mechanical endurance at low temperature 2/ Mechanical endurance at high temperature 1/ Contact resistance Operating characteristics Dielectric withstanding voltage Visual and mechanical examination	2 sample units of each of the following: M8805/93-002 or -003 and M8805/93-027 or -028	
Group V (2 sample units) Electrical endurance sea level inductive load, dc Dielectric withstanding voltage Operating characteristics	1 sample unit of each of the following: M8805/93-002 or -003 and M8805/93-027 or -028	

1/ Test one M8805/93-002 or -003, and one M8805/93-027 or -028.
2/ Same sample units as for low temperature operation.

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TABLE IV. Group B inspection

Examination or test
Group I
Strength of mounting bushing
Thermal shock
Shock
Operating characteristics
Marking visibility
Group II
Salt spray
Marking visibility
Group III
Low temperature operation
Mechanical endurance at low temperature
Contact resistance
Operating characteristics
Group IV
Mechanical endurance at high temperature
Contact resistance
Operating characteristics

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, AV
Navy - AS, MC
Air Force - 19, 99

Preparing activity:
DLA - ES

(Project 5930-1582)

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